Dheeraj Karki

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Education

MSc Robotics **Sep 2026 – Sep 2027 (expected)**
University of Bristol *United Kingdom*

- Current postgraduate study in robotics, robotic systems, machine vision and research methodology.
- Expected completion: 15 September 2027.

M.Tech in Software Systems **Jan 2017 – Jan 2019**
Birla Institute of Technology and Science (BITS Pilani) *India*

- Specialisation: Embedded Systems.
- CGPA: 8.6/10; First Division.
- Dissertation: CAD in Cloud.

B.Tech in Electronics and Communication **Aug 2008 – May 2012**
ICFAI University, Dehradun *India*

- CGPA: 9.42/10; Distinction.
- Thesis: Accelerated Parallel Processing Using OpenCL.
- Study project: Wireless Power Transmission.

Research Interest

Multi-agent robotic systems; decentralised planning and decision-making; reliable information sharing under uncertainty; reinforcement learning; adaptive and trustworthy robotic autonomy.

Robotics and Academic Projects

Beyond Binary Trust: Evidence-Aware Decision-Making *Independent Research | 2026–Present*

- Developed a synthetic multi-source information benchmark to investigate reliability and decision-making under uncertainty.
- Implemented and evaluated majority voting, global reliability weighting and domain-specific reliability weighting using separate calibration and test datasets.
- Investigating evidence quality, provenance, source dependence and justified abstention as extensions to existing trust models.

Mobile Robot Raster-Scanning Project *MSc Robotics | 2026*

- Working with an Arduino-based mobile robot equipped with infrared reflectance sensors and wheel encoders.
- Investigating line-following behaviour, rotation accuracy, wheel drift and positioning during raster-scanning experiments.
- Using sensor measurements, motion control and odometry to investigate practical robotic navigation and repeatability.

Infrared Reflectance Sensor Characterisation *MSc Robotics | 2026*

- Conducted experiments investigating reflectance sensor responses to black and white surfaces, strip width and boundary effects.
- Analysed measurements to investigate sensor footprint and the implications of calibration for reliable robot sensing.

Machine Vision Experiments *MSc Robotics | 2026*

- Practised image filtering, thresholding, morphological operations and object detection using Python and OpenCV.
- Worked with annotated image datasets and investigated machine-learning approaches to visual object and colour identification.

Technical Skills

Programming:	C++, C, Python, JavaScript
Algorithms:	Computational geometry, spatial data structures, planar graph traversal, closed-boundary detection, geometry processing
Robotics:	Arduino-based robot experiments, infrared reflectance sensors, wheel odometry, motion control experiments
Machine Vision:	Python, OpenCV, image filtering, segmentation, thresholding, morphological operations
Software Engineering:	System architecture, performance optimisation, cross-platform development, debugging, code refactoring
CAD and Graphics:	ODA (Open Design Alliance), CAD systems, rendering, interactive graphics

Professional Experience

Arden Software **Jan 2019 – Sep 2026**
Software Development *India / Remote, UK*

Staff Software Developer Sep 2025 – Sep 2026

- Modernised a legacy desktop CAD application through migration to a modern engine, supporting feature parity across web, desktop and cloud platforms.
- Engineered quadtree-based spatial data structures for polygon lookup, point-in-polygon detection and nested polygon-hierarchy resolution, improving geometric query performance by approximately 35%.
- Collaborated across CAD, architecture and UX teams while refactoring core modules for maintainability.

Senior Software Developer Oct 2021 – Sep 2025

- Architected a reusable CAD annotation framework supporting dimensions, text, callouts and leaders.
- Implemented import/export support for DXF, DWG, IPD and internal JSON formats.
- Improved rendering pipelines and geometry-processing functionality for interactive CAD systems.

Software Developer Jan 2019 – Oct 2021

- Optimised HTML5 canvas rendering by intercepting render calls, caching geometry and batching draw operations, reducing rendering latency by approximately 90%.

Graebert GmbH **Feb 2016 – Jan 2019**
Software Engineering *Noida, India*

Senior Software Engineer Jan 2018 – Jan 2019

- Designed extensible plugin frameworks and drawing-property systems for CAD applications.

Software Engineer Feb 2016 – Jan 2018

- Developed custom CAD entities, view systems and interactive editing tools for a cloud-based CAD editor.

Sevya Multimedia Technology **Feb 2014 – Jan 2016**
Software Developer *Hyderabad, India*

- Developed C/C++ automation tools for enterprise clients, including STMicroelectronics and Aristocrat.
- Developed an embedded home-automation prototype for customer demonstrations.

Electronics Corporation of India Limited **Mar 2013 – Jan 2014**
Engineer Consultant *Dehradun, India*

- Led a team of ten operators on a National Population Register project, coordinating daily processing and data-protection procedures.

Other Projects and Ventures

Frido – Productivity Application *Jul 2019 – Jul 2022*

- Co-owned and developed a productivity application for Apple's software ecosystem, collaborating with UI/UX designers.
- Released the application publicly and received recognition on Product Hunt.

Academic Achievements

- Silver Medalist (2nd Rank), B.Tech, ICFAI University, 2012.
- Merit Fellowship Recipient (50% tuition waiver), ICFAI University, 2008–2012.